

# Project Update

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# Agenda

- Progress
  - Project Timeline
  - Design Goal
  - Concept Selection
  - CAD
- Next Steps & Questions

# Project Timeline

- Individually prompted LLM to generate potential work plans given different time constraints
  - 1 week, 2 weeks, 4 weeks, 6 weeks, weekly plans, day-by-day, task breakdowns, etc
- Compared results, used best aspects of each plan to draft a new prompt

| Week | Main goal                                                           |
|------|---------------------------------------------------------------------|
| 1    | Define project scope and select turbine concept.                    |
| 2    | Build the baseline model and establish a runnable simulation setup. |
| 3    | Simulate and compare baseline and alternate designs.                |
| 4    | Refine, validate, and select the final design.                      |
| 5    | Assemble documentation.                                             |

## **Week 1 07/09/2026 to 07/14/2026 - lock scope and choose the concept**

Goals and measurable outcomes for the week:

- Design requirements sheet completed.
- Concept comparison matrix completed.
- One primary concept selected.
- One fallback concept defined or explicitly ruled out.
- Week 2 geometry plan written down.

Manageable tasks:

- Turn the Boston-area site and wind information into a one-page requirements sheet.
- Define the decision criteria: energy capture at low-to-moderate wind speeds, urban suitability, manufacturability, cost realism, and simulation feasibility.
- Compare the candidate VAWT concepts against those criteria.
- Record the reasons each concept is kept or eliminated.
- Choose the primary concept and identify the fallback plan.
- Define the first-pass geometry variables to set in Week 2.

Things the LLM can help with:

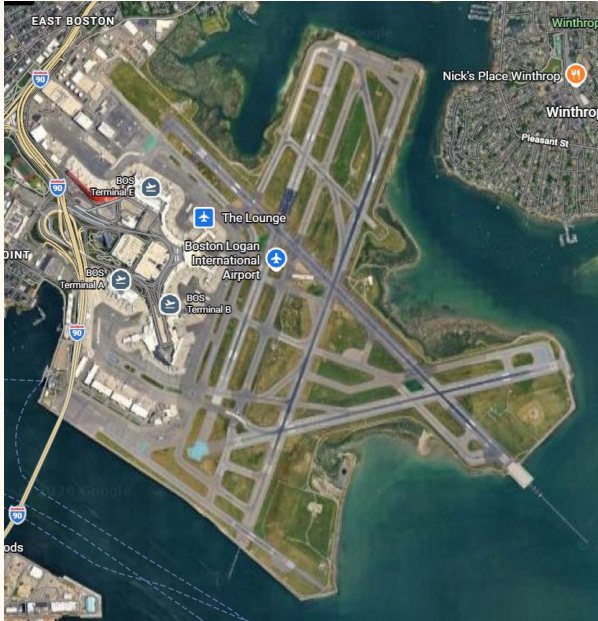
- Turn notes and source material into a requirements summary.
- Build a concept comparison matrix and weighted decision rubric.
- Draft meeting agendas and question lists for the professor and HRI.
- Help document why the selected concept fits the project goals better than the rejected ones.

# Reflections

- Hard to get diverse results without drastically changing the prompt
- Some of the details it filled in felt like filler as opposed to a useful breakdown
- It seemed almost unnecessary to have the LLM do this, since we are already pretty good at knowing our work styles and how to break a project into tasks
- Liked the list of things the LLM can help with
- Useful for laying out a detailed plan in advance when we are still learning about the process and don't know enough to break that task down (eg CFD)
  - we don't always need a detailed breakdown in advance since the timeline may change before then

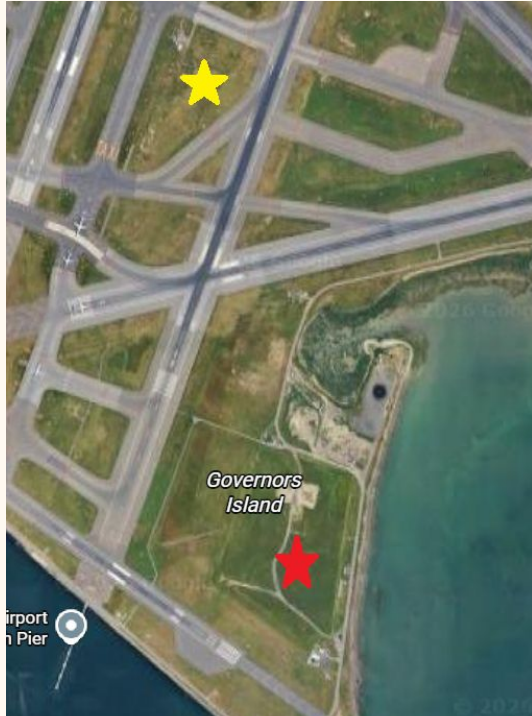
# Design Goal

Proposed location: Governor's Island, Boston Logan Intl. Airport



- Abundantly available wind data with very good resolution
- Avg wind speed  $\sim 5$  m/s
- High energy demand
- Demonstrated interest in reducing carbon emissions

# Design Goal



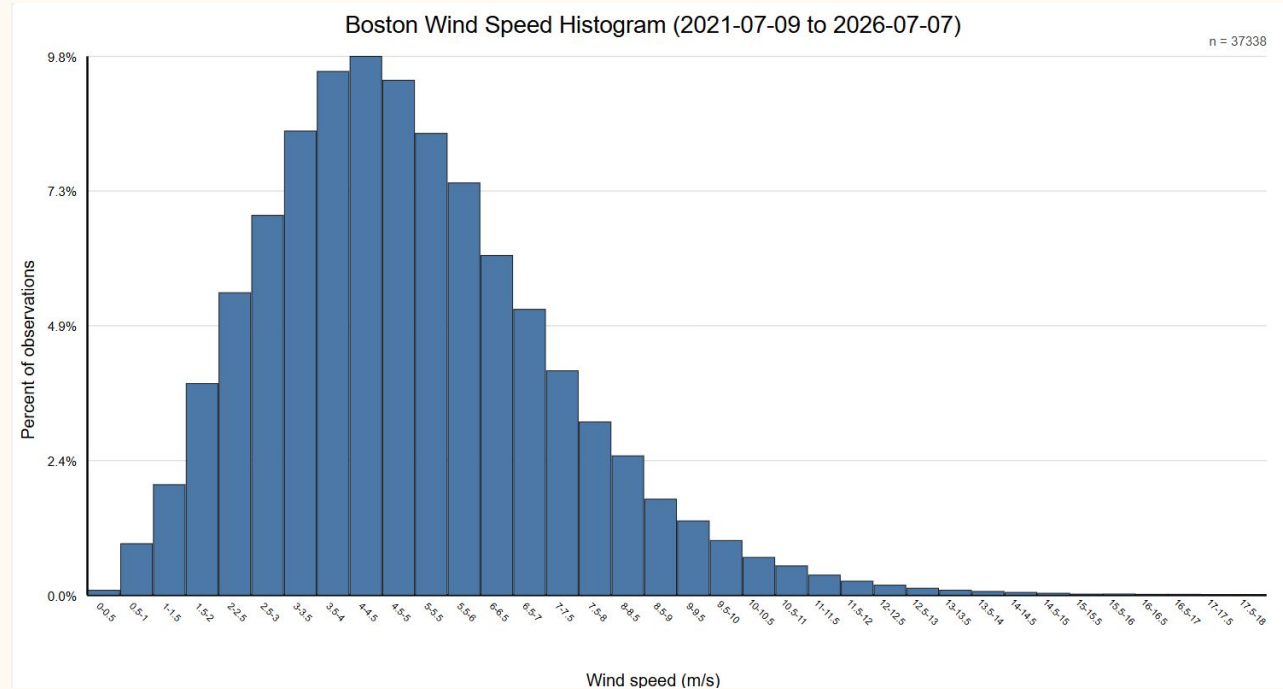
- Wind data collected at 9 m high, center of airfield (yellow star)
- Proposed turbine center rotor height 9 m (total height ~ 10 m)
- FAA regulations require distance of ~200 m from runway
- Proposed turbine location on Governors Island (red star)
  - ~ 10 acres of non-restricted space

# Design Criteria

- Optimized for installation site
  - Low cut in speed to take advantage of available wind power
  - Rated speed and cut out speed matching BOS wind conditions
- Economic feasibility
  - Goal: offset its cost within half its lifetime
- Comply with airport regulations
- Not more complex than necessary
- Minimal maintenance requirements
- Feasible for our project timeline

# Concept Selection

Cut-in: < 3 m/s  
Rated: 7-10 m/s  
Cut-out: > 15 m/s

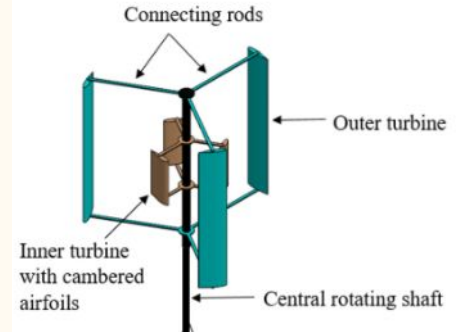
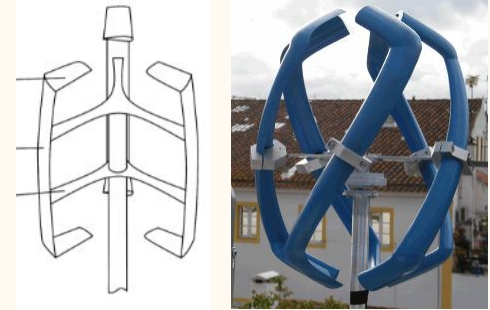


# Concept Selection

- Individually selected 8-10 designs to consider pursuing
  - Implemented LLM wiki to sort through 63 possible designs and discuss pros and cons for BOS
- Selected 6 non-overlapping designs and most important criteria
- Personally ranked designs, then compared to LLM rankings (quantitative)
- Weighted criteria and scored the designs
  - human weights, series of LLM weights based on personas
    - Beginner engineer
    - Expert engineer
    - Best for BOS

# Concept Selection

| Rank | Candidate                                                   | Weighted score (/5) | Notes                                                                                                                                 |
|------|-------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Self-starting straight-bladed H-rotor Darrieus              | 4.40                | Highest overall mainly because it stays strong in both startup and efficiency, which dominate your weighting.                         |
| 2    | <a href="#">vj20 Proposed Hybrid VAWT</a>                   | 4.25                | Very close second because of its top efficiency score and strong startup score, now nearly matching the straight-bladed H-rotor path. |
| 3    | <a href="#">vj9 Scooplet-Based Savonius</a>                 | 3.08                | CAD score helps a lot here, but low efficiency and rated-speed fit keep it mid-pack.                                                  |
| 4    | <a href="#">va20 Involute Rotor with Wind Flow Modifier</a> | 3.05                | Good efficiency and decent growth potential, but weak rated-speed fit and low CAD score reduce the total.                             |
| 5    | Low-TSR helical Darrieus                                    | 2.88                | Good rated-speed fit, but the lower startup and efficiency scores hurt because those categories carry most of the weight.             |
| 6    | <a href="#">va23 50% STS-VAWT</a>                           | 2.48                | The weak startup score hurts most under this weighting, even though CAD and cut-out fit are better.                                   |



# Concept Selection

- Checked validity of initial design selection
  - Ranked all H-type VAWTS
  - Ranked all 63 VAWT designs in the wiki
- Repeatedly saw the same top 2 turbine designs
- Decided to move both forward into CAD

## EN0005 vs vj20 Across Weightings

This compares [\[\[va9 EN0005 Self-start Darrieus VAWT\]\]](#) and [\[\[vj20 Proposed Hybrid VAWT\]\]](#) across the different weighting prompts currently used in this worksheet.

| Weighting prompt                                                                                                      | EN0005 score (/5) | vj20 score (/5) | Higher-ranked design                                | Why the result shifts                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------|-------------------|-----------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Performance-first concept weighting: startup 35%, efficiency 35%, CAD 15%, rated speed 10%, growth 2.5%, cut-out 2.5% | 4.30              | 4.25            | <a href="#">va9 EN0005 Self-start Darrieus VAWT</a> | EN0005 wins slightly because its startup and cut-out evidence are stronger, while vj20 wins on efficiency.                                                                 |
| BOS-focused all-design weighting: startup 30%, efficiency 25%, rated speed 15%, CAD 15%, growth 10%, cut-out 5%       | 4.30              | 4.15            | <a href="#">va9 EN0005 Self-start Darrieus VAWT</a> | EN0005 still wins because BOS-fit rewards low-speed startup and proven high-wind stability more than raw peak Cp.                                                          |
| Beginner-friendly all-design weighting: CAD 30%, growth 25%, startup 20%, efficiency 15%, rated speed 5%, cut-out 5%  | 4.45              | 4.15            | <a href="#">va9 EN0005 Self-start Darrieus VAWT</a> | EN0005 opens the biggest gap because it combines strong startup with a simpler-looking straight-body geometry, while vj20 still carries a more involved dual-rotor layout. |
| Expert BOS performance weighting: startup 20%, efficiency 25%, rated speed 20%, cut-out 10%, growth 20%, CAD 5%       | 4.13              | 4.28            | <a href="#">vj20 Proposed Hybrid VAWT</a>           | vj20 finally moves ahead when optimization headroom, rated-speed fit, and peak aerodynamic performance are weighted more heavily than simplicity.                          |

# Reflections

- LLM wiki *drastically* reduced the time needed for concept selection
- Very helpful to instantly pull out specific information across our ~60 sources and compare findings from many studies
  - Chat interface makes it much easier to find what you need even if you have limited knowledge about the content
- Removed the opportunity for deep research and learning about the vawt concepts and designs
  - No extended time to think about the options, decision was made in a day
- Results feel a little bit arbitrary and because of our own limited knowledge we can't feel entirely confident about the decisions



Code Editor



```
innerRotorBottomZ, global = true)
24 |> rotate(axis = [0, 0, 1.0], angle =
    innerPhaseAngle, global = true)
25 |innerBlades = patternCircular3d(
    innerBladeSeed,
26 |instances = innerBladeCount,
    axis = Z,
27 |center = [0m, 0m, 0m],
    arcDegrees = 360deg,
28 |rotateDuplicates = true,
    )
29 |
30 |
31 |
32 |
33 |
34 |shaftSketch = sketch(on = offsetPlane(XY,
    offset = shaftBottomZ)) {
35 |  shaftCircle = circle(start = [var 0.035m,
    var 0m], center = [var 0m, var 0m])
36 |
37 |  coincident([shaftCircle.center, ORIGIN])
38 |  diameter(shaftCircle) == shaftDiameter
39 |  horizontal([shaftCircle.center,
    shaftCircle.start])
40 | }
41 |shaftRegion = region(point = [0m, 0m], sketch
    = shaftSketch)
42 |centralShaft = extrude(shaftRegion, length =
    shaftLength)
43 |
```

Project Files



innerBladeDu06W200.kcl

main.kcl

outerBladeNaca0018.kcl

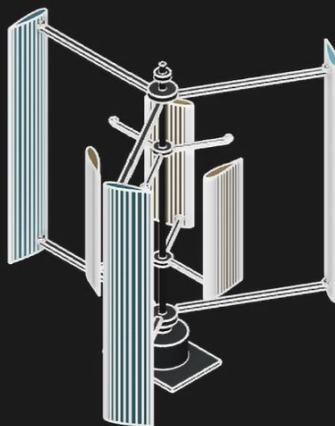
parameters.kcl



Start Sketch



Angularity



Zookeeper

thick.

- Outer blade joint pads: 120 mm diameter × 30 mm thick.
- Inner blade joint pads: 90 mm diameter × 25 mm thick.
- Fastener heads: 30 mm diameter × 12 mm high.
- Bearing housings: 180 mm OD × 76 mm bore × 70 mm thick.
- Generator: 560 mm OD × 86 mm bore × 320 mm high.
- Base plate: 1050 × 750 × 45 mm.
- Battery pack: 420 × 220 × 160 mm, offset 390 mm in X and -210 mm in Y.
- Battery tray: battery footprint plus 40 mm, 20 mm thick.

Possible next edits:

1. Increase the airfoil polyline density to use the full DAT point sets.
2. Add a cutaway/ghosted view option so the inner rotor is always visible.
3. Add a simple exploded assembly variant for clearer documentation.



Thoughtful ^



Zookeeper can make mistakes. We send selection context to help. Always verify information.



v1.3.5

zoo.dev



Settings



No selection

All



m



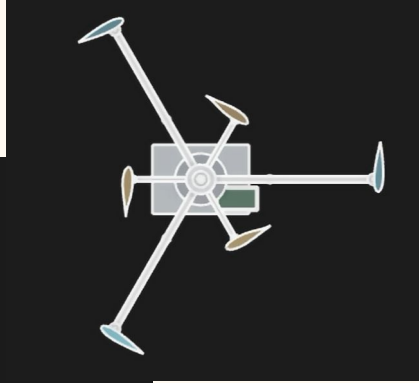
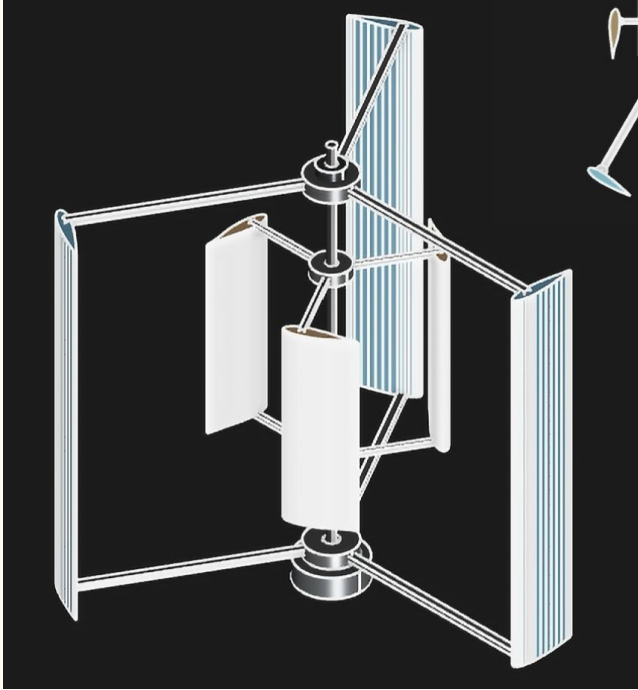
# Process + Reflections

- Used LLM wiki to find design specs and generate a zookeeper prompt
  - Incredibly long and detailed, but suggested a shorter version as a next step
- Uploaded image of prototype, blade profile, and a sketch
- One by one ask for edits to design
  - Shorter prompts were best
  - Started asking it to report what edits it made to the code and how I could modify them
- Shortened version of the long prompt generated a mostly correct model with some large flaws like disconnected blades (9 min)
- “Short production-ready” prompt from LLM generated a completely correct basic model (15 min)

Because this design didn't have specific geometry information it was challenging to get Zookeeper to model it

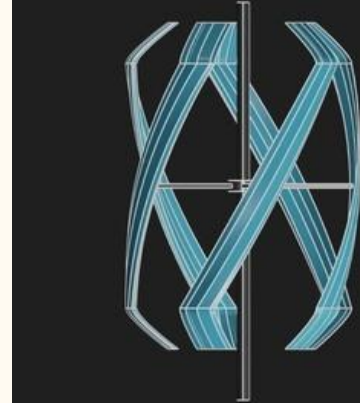
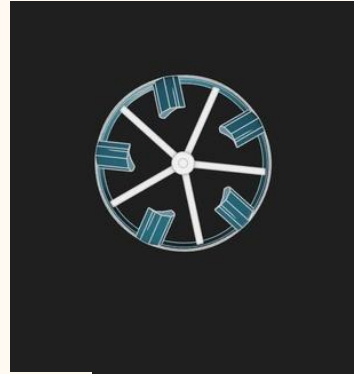
Going forward, making small edits proved to be difficult without understanding the root cause of the issues I was trying to fix

# CAD



← Hybrid H-type (vj20)

EN0005 H-type (va9) →



# Design Selection Conclusion

- Hybrid H type was easily modeled and has very promising performance specs
  - Always ranked in top designs during selection process
- EN0005 H type was more challenging to model
- EN0005 was frequently winning comparisons due to low cut in speed
  - Difficult to test in SimScale
  - Potentially difficult to recreate w/out design specs
  - Not necessarily the most important spec for BOS site
- Initial thought: combine design strengths by applying EN005 H type unique geometry to the Hybrid H type turbine
  - Airfoil
  - Blade end positions

# Next Steps

- Continue learning about CFD and how to use SimScale
- Complete validation studies on an airfoil and a simple VAWT
- Complete baseline CAD drawing of hybrid design

# Questions

- How to test or estimate cut-in speed virtually?
  - QBlade?
- Process for selecting independent variables to test?
  - DOE